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Roll No

CE-8004(4) (CBGS)

B.E. VIII Semester Examination, June 2020

Choice Based Grading System (CBGS)

Finite Element Method

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Discuss the advantages and disadvantages of FEM over. 7
i) Classical Method
ii) Finite Difference Method
b) Explain the concept of FEM briefly and outline the procedure. 7

2. Eigen Value Analysis. Consider the differential equation. 14

$$\frac{\partial u}{\partial t} - \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = F \text{ in a square region, subjected to boundary conditions:}$$

$$\frac{\partial u}{\partial x}(0, y, t) = 0 ; \frac{\partial u}{\partial y}(x, 0, t) = 0 ;$$

$$u(x, 1, t) = 0 ; u(1, y, t) = 0 \text{ and initial condition}$$

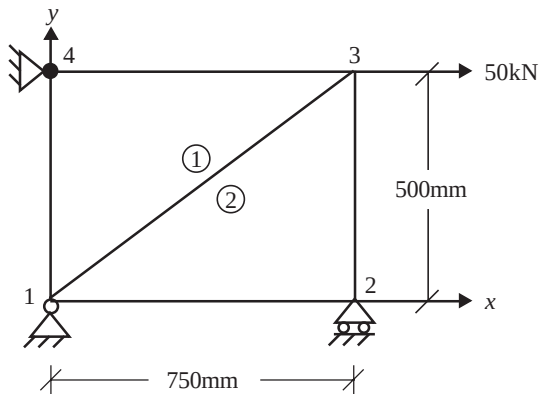
$$u(x, y, 0) = 0$$

3. Explain the following terms clearly: 14
i) Local coordinates
ii) Global coordinates
iii) Natural coordinates
iv) Area coordinates
4. Determine the shape function for a three noded bar element using Lagrangian Functions. Plot their shapes. 14

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5. Find the nodal displacement and element stresses in the propped beam shown below. Idealize the beam into two CST elements as shown. Assume plane stress condition. 14

Take $\mu = 0.25$ $E = 2 \times 10^5 \text{ N/mm}^2$, Thickness = 15 mm.



6. How the matrix displacement equation can be solved using Gaussian Elimination Method. Explain step wise. 14
7. Explain the term 'Shape Functions'. Why polynomial terms are preferred for shape functions in finite element method? 14
8. For the three-bar truss shown in fig (8.1). Determine the Nodal Displacements and the stress in each member. Find support reaction also. Take $E = 200 \text{ GPa}$. 14

